

Query Solutions

Query 1: Average Doctors' Salary per Department

γ dname, AVG(salary) (DOCTORS)

Explanation: Uses grouping operator to calculate average salary grouped by department name.

Query 2: Patients Operated on by "Dr. Smith" on "10-23-2022"

```
 $\pi$  PATIENTS.name (  
     $\sigma$  DOCTORS.name='Dr. Smith'  $\wedge$  OPERATIONS.date='10-23-2022' (  
        DOCTORS  $\bowtie$  DOCTORS.ID=OPERATIONS.DID OPERATIONS  $\bowtie$   
        OPERATIONS.Pnum=PATIENTS.Pnum PATIENTS  
    )  
)
```

Explanation:

- Selects Dr. Smith from DOCTORS
 - Filters operations for the specific date
 - Joins DOCTORS $\rightarrow$ OPERATIONS $\rightarrow$ PATIENTS
 - Projects patient names
-

Query 3: Doctors Who Have No Patients

```
 $\pi$  name (DOCTORS) -  $\pi$  DOCTORS.name (DOCTORS  $\bowtie$  ID=DID OPERATIONS  $\bowtie$   
Pnum=Pnum PATIENTS)
```

Explanation:

- Gets all doctor names
 - Finds doctors who have performed operations on patients
 - Uses set difference to find doctors with no patients
-

Exercise 2: SQL Trigger (50 points)

Trigger Name: log_student_promotion

Trigger Logic:

```
CREATE TRIGGER log_student_promotion
AFTER UPDATE ON student_mast
FOR EACH ROW
BEGIN
    IF OLD.ST_CLASS != NEW.ST_CLASS THEN
        INSERT INTO stu_log (description)
        VALUES (CONCAT(NEW.STUDENT_ID, ' | student promoted from class ',
                        OLD.ST_CLASS, ' to ', NEW.ST_CLASS));
    END IF;
END
```